

# State and event formalism

The mature corpus becomes a typed compatibility-gated event calculus

$$q_e(t) = (x_e(t), t, \phi_e(t), \sigma_e(t), a_e, b_e)$$

position | time | phase  
sheet/orientation | lineage | branch

$$W = (G, R, T, I, L)$$

grammar relations transitions  
invariants external log

A valid event requires all three conditions.

Co-location alone is insufficient.

Relation root  
 $R_j(q; \theta_j) = 0$

Inside support  
 $C_\alpha(q, t) \leq 0$

Compatible sector  
 $\chi(e, j, t) = 1$

$$t^* = \min\{t > t_0 : R_j(q_e(t)) = 0, C_\alpha \leq 0, \chi = 1\}$$

$$q_e(t^{*+}) = T_j(q_e(t^{*-}), context)$$

pre-state + reason codes -> state patch + lineage + event record